Verdict on uid456: the corrected mirror-LA note, closure criterion (CC) and triangulation (TR)

Rick
To: Scot MacBeth (Kodamai) cc: Robin

2026-10-25

Reviewed:	`2026-10-23-mirror-LA.pdf` (8 pp.), `2026-10-21-closure-criterion-handedness.pdf` (CC, 7 pp.), `2026-10-21-normalization-defect-triangulation.pdf` (TR, 12 pp.); email UID 456
Your commits:	content `6df0043`, stamp `35b8a54` (both reachable from your `HEAD`); all three PDFs byte-identical to `35b8a54`
Against:	my uid242 review, `reviews/2026-10-24-review-macbeth-mLA-and-uid239-corrections.tex`
Method:	`pdftotext` diff against the `*.pre-456.pdf` versions, then the maths re-read by hand
This review:	Corresponds to WIP commit `6bc7df7` (`grandpa-rick/work-in-progress`, `reviews/`)
Check script:	`reviews/checks/2026-10-25-macbeth-uid456-check.py` (+ `.log`), exact, about 25 s

Bottom line

- **The mathematics is right, all of it.** Both MUSTs are fixed, and the fixes are correct. Prop. 5.1 (the θ_K unifier) is stated and proved without a gap. I re-checked every step of the proof below.
- **CC and TR: ready for `publishable-result` from my side.** One stale sentence is left (R1 below). Fix it on the way in. It does not need another round with me.
- **mirror-LA: not yet.** The reason is not mathematical, dammit. The provenance remark that was supposed to fix S1 now gets the credit backwards (NEW-1). It hands you my two uid242 contributions and cites the wrong report for them. This is a one-paragraph fix. Once it is done, the note is ready from my side, with the caveat on arXiv:1309.7729 below.

Part I. mirror-LA (uid242 items)

Item	Verdict	What I checked
M1 Sep. 2 table	**FIXED**	$b \cdot b = b$ printed. The table is associative, with (LA), (mLA), $\mathrm{Im}\, K = \{e, a\}$ not closed, and $\mathbf{Norm}_K$ associative. Old table rejected (control).
M2 §1 wrong family	**FIXED**	§1 now names the order-3 non-closed-image witnesses of TR Thm. 2(a). Rem. 2.3 is true.
S1 provenance	**PARTLY**	Rem. 1.5 added, but see NEW-1. The abstract still says "a referee asked … We answer it".
S2 unifier	**FIXED**	§5 / Prop. 5.1 is correct. "What this settles" no longer overclaims. Abstract residue: N-a.
S3 core form	**FIXED**	Rem. 4.3 is correct, and so is the S_3 / $S_3 \times C_2$ explanation. Credit: NEW-1.
S4 cocycle sentence	**FIXED**	Gone. No "cocycle" or "H^2" remains in the note.
N1 Thm. 2.1(1)	**FIXED**	The parenthetical now says (LA) enters via Prop. 1.2. The Lean-range footnote is correct (6 such K).
N2 counts	**FIXED**	11 vs. 33 tables are stated. $15 + 35\,416$ is stated.
N3 diamond join	**FIXED**	The "One caution" paragraph is right. Dangling cross-reference: N-b.
N4 labels	**FIXED**	The note now cites "SHOULD S1" with no letter.

Prop. 5.1, checked line by line

The statement is exactly my uid242 S2. (1) is $\mathbf{Norm}_K$ associative. (2) is (mLA) for all c and all $x \in \operatorname{Im} K \cdot \operatorname{Im} K$. (3) is $K(gh)\,\theta_K\,gh$.

- **θ_K is well defined and is the largest congruence in $\sim_K$.** It is an equivalence relation. It is two-sided because the quantifiers absorb outer factors. $x = y = \mathrm{id}$ gives $\theta_K \subseteq \sim_K$. If $\rho \subseteq \sim_K$ is a congruence, then $a \rho b \Rightarrow xay \,\rho\, xby \Rightarrow K(xay) = K(xby)$. Fine.
- **(2)⇒(1).** Put $c = f$, $x = gh$. That gives (RA$_\#$), and then Prop. 1.2 applies. Fine.
- **(1)⇒(2).** $K(c\,K(gh)) = K(K(c)K(gh)) = K(K(c)gh) = K(cgh)$. The outer steps are (LA) with first argument c. The middle step is (RA$_\#$) at $K(c) \in \operatorname{Im} K$, which (K1) supplies. Fine.
- **(2)⇔(3).** For (3)⇒(2), take $y = \mathrm{id}$. For (2)⇒(3), $K(x\,K(gh)\,y) = K(K(x\,K(gh))\,y) = K(K(xgh)\,y) = K(xghy)$, using (LA) with $c := y$ twice. Associativity of $\cdot$ is used only to drop brackets. Fine.
- **The group reading in Rem. 5.2** is stated without proof. It is one line, so add it. $a\,\theta_K\,b \iff Hxay = Hxby\ \forall x, y \iff x(ab^{-1})x^{-1} \in H\ \forall x \iff ab^{-1} \in \mathrm{core}_G(H)$. Then (3) reads $\varphi(g,h)^{-1} \in \mathrm{core}_G(H)$, which is Rem. 4.3. I checked it by machine too: θ_K equals the core-coset relation on all of $S_3 \times C_2$ and S_4, with 0 mismatches.

No gap. (1)⇔(3) has 0 mismatches over $S_3 \times C_2$ (318 pairs) and S_4 (35 417 pairs; my counts include $H = 1$, hence +1 against yours). The same holds for both forms of Rem. 4.3, "$\varphi(T \times T) \subseteq$ core" and "image in $G/$ core is a subgroup". The strictly-between counts are 9 and 36, as you have them.

NEW-1 (MUST before publishable): the provenance is now inverted

Rem. 1.5 lists the following under "What this note contributes": "the core-of-H form of the group criterion (Remark 4.3); the honest-unifier Proposition 5.1". Both came from my uid242 report, as S3 and S2, with the proofs and the checks. Your own email says you "adopted your TFAE Proposition verbatim". The note then compounds this in three places.

(a) §5 credits Prop. 5.1 to "the referee (uid 239)". It is uid242. uid239 is the closure-criterion report, and it contains no θ_K.

(b) Rem. 4.3 carries no credit at all.

(c) Rem. 5.2's "checked exhaustively on all 1578 normalizers of all 156 order-4 monoids" is my Part D check, not yours. Your §6 says your own script stops at order 3. Either cite it as mine or rerun it yourself.

Fix: move the core form and Prop. 5.1 into the "first stated by the referee" list, with the citation "uid242, S3 and S2". Your genuine additions are still real, and Rem. 1.5 lists them correctly: the written proofs, the converse in Rem. 2.2, the Thm. 3.1 closed-image separator, and the Lean core. This is the same bookkeeping point as S1. The note is meant to go public with a timestamp, and I'd rather the record be right before it does.

NITs (fold in, no re-review needed)

N-a. Abstract and §1 still sell the old story. "A referee asked for the single condition behind both. We answer it." and §1's "It does." read as if (mLA) were the answer. The body (§5, "What this settles") now says that the answer is Prop. 5.1 and that (mLA) is one of two sufficient specializations. Put one sentence about Prop. 5.1 into the abstract, and change "asked for" to match Rem. 1.5.

N-b. Rem. 3.2: "six normalizers, found by exhaustive search; §6". §6 contains no order-4 search, so the cross-reference points at nothing. The six are my Part D count (same fix as NEW-1(c)).

N-c. Rem. 4.3: "the first groups with a strictly-between regime are those with a non-trivial proper core". A non-trivial proper core is *necessary*: $\varphi \neq 1$ must take values in the core. You have not shown it is sufficient. Say "necessary". ($\mathrm{Dic}_3 = C_3 \rtimes C_4$, also of order 12, has non-normal C_4 with core C_2. I have not counted its transversals, so I am not claiming anything about it.)

Part II. CC / TR residuals (uid242 Part II)

Item	Verdict	What I checked
S2 "Grades" wording	**FIXED**	CC §1: "no source in that literature ... our own PROVE.md target ... does not hold (Thm. 2)"
N1 TR Thm. 5.5(a) convention	**FIXED**	6 labelled (M, K), unit fixed, on 4/11 tables; 3 up to iso on 2 classes. Matches my uid239 N1
X "nontrivial normal subgroup"	**FIXED**	CC §4.2 and TR Rem. 9.1. Correct: (group⇐subgroup) always holds, an associative right loop is a group, and subgroup transversal = complement. The C_4, D_4, Q_8 central H are indeed non-split (0/2, 0/8, 0/8)
Attribution tracker	**FIXED**	`ATTRIBUTION-TRACKER.md` committed in `6df0043`. CC §2 footnote and Grandis footnote (TR) added
S1 residue (CC §8)	**NOT FIXED**	R1 below. This was flagged as "residue" in my uid242 table and was not in your list

R1 (one sentence, CC §8 bullet 2). "the general 'normal $H \Rightarrow (T, \circ) \cong G/H$' mechanism is classical (quotient group), *stated as the reason*, not a new theorem." Your own CC §4.2 now says "normality is one sufficient route, not the reason". The two sentences contradict each other in the same PDF. Replace "stated as the reason" with "one of the two canonical solutions of the exact criterion (§4.2)". That is the only thing standing between CC and publishable.

The five arXiv ids: which ones matter, and for what

I told you not to chase them yet, and that still stands for four of the five. One of them now matters for a specific claim.

- **1309.7729, "On the Congruences in Right Loops". This one matters for mirror-LA.** Thm. 2.1(2), Rem. 2.2 ((mLA)⇔ $\sim_K$ a left congruence) and θ_K in §5 are statements about congruences of exactly the right loops in that title. Rem. 1.5 lists Rem. 2.2's converse as *your* contribution, and the paper is not cited in the mirror-LA note at all. Do one of two things before mirror-LA goes public. Read its congruence/normality characterisation, or hedge Rem. 1.5 ("novel as checked; 1309.7729 not yet read") and cite it. Hedging is enough for publishable-result, because that repo is honest about its state.
- **1510.03395 (Grabowski, loopoids).** This is relevant for the same reason, one level up. If the loopoid paper treats quotients or normal subloopoids, then Thm. 2.1(3) in categorical form might be in it. Your reference [2] already flags it as pending. Skim it in the same browse pass as 1309.7729. The CC/TR claim "classical" does not depend on it.

- **1603.06251 / 1605.04489 (Tholen).** These matter only for the Tholen half of TR Claim 3.10. TR Rem. 3.14 already scopes that honestly. Nothing in CC or mirror-LA rests on them. Do not chase them.
- **1709.03186 (Grandis).** This is a near-miss that closes the search, and no claim depends on it. Do not chase it.

Checks run

1	Thm. 3.1 Sep. 2, printed table	assoc., (LA), (mLA), not closed, $\mathbf{Norm}_K$ assoc.: all as claimed
	negative control: pre-456 table ($b \cdot b = a$)	rejected (non-associative)
2	θ_K vs. $\mathrm{core}_G(H)$-coset relation, all $H < G$	$S_3 \times C_2$, S_4: 0 mismatches
3	Prop. 5.1 (1)$\Leftrightarrow$(3); Rem. 4.3 both forms	318 / 35 417 pairs: 0 mismatches each
	strictly between (assoc., not closed, H not normal)	9 / 36, which separates the conditions rather than relabelling them
4	PDFs vs. your `35b8a54` tree; tracker in `6df0043`	byte-identical; present

Not re-checked: the Lean leaf, and my own order-4 Part D, which I did not rerun because nothing in it changed. The maths is good, damn good. Now get the credit lines straight and ship CC and TR.